MCBB 3.0 AWS LAB-POC

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1. AWS Cloud WAN Terminology & Concepts

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Quick reference for key AWS Cloud WAN terms and policy constructs used throughout this lab build.

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1.1 Core Components

Term	Definition
Core Network	The global C-WAN resource that spans all regions; contains the policy document, segments, and attachments.
Core Network Edge (CNE)	A regional presence of the core network. Each AWS region in your policy gets a CNE that handles routing and forwarding for that region.
Segment	An isolated routing domain within C-WAN, similar to a VRF. Attachments in the same segment can communicate; different segments cannot unless explicitly peered.
Segment Peering	Allows route exchange between two segments within the same CNE. Used to enable Shared Services to reach both Prod and Non-Prod.
Attachment	A connection between a resource (VPC, DXGW, S2S VPN, TGW, or Connect) and the core network. Each attachment belongs to exactly one segment.
Attachment Policy	Rules that automatically map new attachments to segments based on tag conditions. Evaluated in order; first match wins.
Network Route	A route entry in a segment's route table at a specific CNE, showing destination prefix, attachment, and route type (propagated or static).

1.2 Policy Actions & Routing Constructs

Term	Definition
Segment Actions	Policy rules that define how traffic is routed between segments and attachments. Includes share-with, send-to, and send-via actions.
Send To	Forwards traffic destined for a specific segment to an intermediary attachment (e.g., Inspection VPC) before delivery. Used for single-hop inspection.
Send Via	Inserts a middlebox (e.g., firewall) into the traffic path between two segments. Supports both single-hop and dual-hop inspection patterns.
Single Hop	Traffic passes through one intermediary attachment before reaching destination. Example: Prod VPC Inspection VPC On-Prem via DXGW.
Dual Hop	Traffic passes through two intermediary attachments before reaching destination. Example: Source VPC Inspection (region A) Inspection (region B) Dest VPC.
Share With	Shares routes from one segment with another without full peering. Provides one-directional or selective route sharing.
When (Condition)	A condition block in the attachment policy that matches on resource tags (e.g., tag:Environment = Prod) to determine segment placement.
Edge Override	Allows per-region overrides of the global segment/attachment policy. Used when a specific CNE needs different routing behavior.

2. LAB Overview & Objectives

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2.1 Purpose

The MCBB 3.0 AWS Lab environment serves as a pre-production validation platform to prove out the Cloud WAN with Split DXGW architecture defined in the MCBB 3.0 production design document (v.25, Jan 2026). The lab mirrors the production topology at a reduced scale, enabling the team to validate design requirements (DRs), test failure scenarios, refine Terraform modules, and document operational runbooks before production deployment.

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2.2 Scope

- AWS Cloud WAN (C-WAN) core network across three lab regions: us-east-1, ap-southeast-2, us-west-2
- C-WAN segmentation: Prod, Non-Prod, Shared Services (peered to Prod + Non-Prod), and IBB segments
- Cloud Network Edge (CNE) attachments per region
- Split DXGW architecture with regional DXGW instances
- Multiple VPC types per region: Prod, Non-Prod, Inspection, and Shared Services
- Inspection VPC for firewall deployment (AWS Network Firewall or third-party appliance)
- Direct Connect integration via Megaport virtual cross-connects (VXCs)
- BGP route propagation validation with community tagging
- Cross-region traffic flow validation over C-WAN backbone
- International Backbone (IBB) overlay design
- Service chaining through inspection VPCs for east-west and north-south traffic
- Infrastructure as Code (Terraform Enterprise) with GitLab CI/CD runners
- Network observability: interim CloudWatch/VPC Flow Logs, target state ThousandEyes + Splunk + Selector AI

2.3 Out of Scope

- Full-scale 100GE Cloud OnRamp capacity testing
- LATAM region deployment (Argentina, Panama, Mexico)
- Inter-CSP connectivity (Azure, GCP integration points)
- QoS enforcement at egress (DR 4.5 — not supported by CSP)
- Versa SD-WAN / SDR overlay — removed from design due to operational issues

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2.4 Key Design Change: Versa Removal

The original MCBB 3.0 design specified Versa SDRs for the International Backbone (IBB) SD-WAN overlay. For AWS Cloud Inter/intra regional connectivity due to recent operational issues with Versa, the overlay between regions connectivity will be re-architected using cloud-native CNE (CloudWAN Network Edge):

- AWS Cloud WAN native intra/cross-region routing
- Public VIF over existing DX connections for IBB underlay (as referenced in production design doc)

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2.5 Inspection VPC Architecture

Each region includes a dedicated Inspection VPC for centralized security appliances supporting east-west inspection (inter-VPC traffic routed through firewall before reaching destination), north-south inspection (traffic to on-prem/internet inspected before leaving the region), and service chaining via C-WAN routing policies (send-to / send-via actions).

Firewall options under evaluation:

- AWS Network Firewall — cloud-native, integrates with VPC route tables
- Third-party appliance firewall (e.g., Palo Alto, CheckPoint, Fortinet) — deployed as EC2-based NVAs

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3. LAB Topology & Network Diagrams

3.1 Lab Region Selection

Lab Region	AWS Region	DXGW Scope	VPCs Deployed
US East	us-east-1 (N. Virginia)	dxgw-us	Prod, Non-Prod, Inspection, Shared Svc
APAC	ap-southeast-2 (Sydney)	dxgw-apac	Prod, Inspection
US West	us-west-2 (Oregon)	dxgw-emea	Prod, Non-Prod, Inspection

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3.2 VPC Architecture Per Region

us-east-1 (N. Virginia)

VPC Name	CIDR FOR INITIAL BUILD	Purpose	C-WAN Segment
lab-prod-use1	10.20.0.0/21	Production workloads	Prod
lab-nonprod-use1	10.21.0.0/21	Non-production / dev / staging	Non-Prod
lab-inspection-use1	10.22.0.0/22	Firewall appliances (AWS NFW or 3rd party NVA)	Inspection attachment
lab-shared-use1	TBD	Shared services (DNS, logging, tooling)	Shared Services (peers to Prod + Non-Prod)

ap-southeast-2 (Sydney)

VPC Name	CIDR	Purpose	C-WAN Segment
lab-prod-apse2	10.30.0.0/21	Production workloads	Prod
lab-inspection-apse2	10.32.0.0/22	Firewall appliances (AWS NFW or 3rd party NVA)	Inspection attachment

us-west-2 (Oregon) (To be Used for MnA/Client integration Simulation/DXes failover)

VPC Name	CIDR	Purpose	C-WAN Segment
lab-prod-usw2	TBD	Production workloads	Prod
lab-nonprod-usw2	TBD	Non-production / dev / staging	Non-Prod

lab-inspection-usw2	TBD	Firewall appliances (AWS NFW or 3rd party NVA)	Inspection attachment

Note: us-west-2 VPC CIDRs must be from a different address range than us-east-1 (10.20.x.x) and Sydney (10.30.x.x) to avoid overlap.

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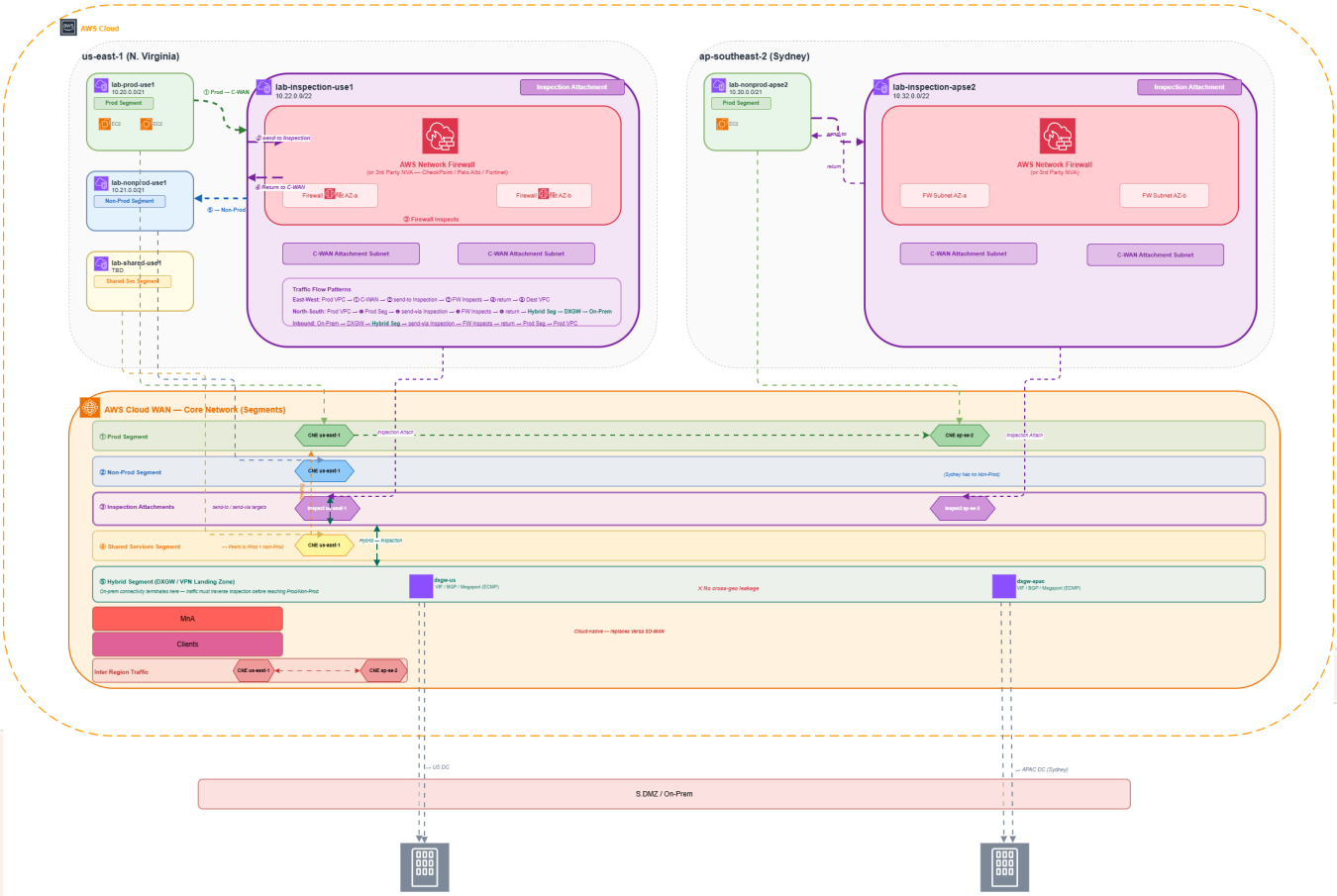
3.3 High-Level Lab Topology

- C-WAN Core Network with four segments: Prod, Non-Prod, Shared Services (peered to Prod + Non-Prod)
- One CNE per region attached to the core network
- Multiple VPCs per region attached to appropriate C-WAN segments
- Inspection VPC per region with dedicated C-WAN attachment for service chaining
- Two regional DXGW instances: dxgw-us, dxgw-apac
- Megaport VXCs delivering hosted DX connections to each DXGW

3.4 Low Level Lab Architecture Diagram

A single comprehensive low-level diagram covering the full lab architecture: C-WAN core network with all 8 segments (Prod, Non-Prod, Shared Services, Hybrid, Clients & MnA), CNEs per region, VPC attachments with segment mapping, Inspection VPCs with firewall placement and service chaining flow (send-to/send-via), split DXGW instances with DX /Megaport connections, route propagation paths (CNE-to-CNE and CNE-to-DXGW), cross-region traffic flows, and IBB routing for international traffic, which will transverse via the CNE edges on AWS backbone.

MCBB 3.0 — AWS Cloud WAN Lab Architecture



VPC & Segment Route Tables

us-east-1 Route Tables

lab-prod-use1

Destination	Target	Notes
10.20.0.0/21	local	VPC local
0.0.0.0/0	cwan-att	Non-Prod, Sydney, On-p (via Inspection)

lab-nonprod-use1

Destination	Target	Notes
10.21.0.0/21	local	VPC local
0.0.0.0/0	cwan-att	

Destination	Target	Notes
TBD (Shared Svc)	local	VPC local
10.20.0.0/21	cwan-att	Prod (via segment peering)
10.21.0.0/21	cwan-att	Non-Prod (via segment peering)

lab-inspection-use1

Firewall Subnet Route Table

Destination	Target	Notes
10.22.0.0/22	local	VPC local
0.0.0.0/0	NAT-GI	Return inspected traffic → C-WAN

C-WAN Attachment Subnet Route Table 1

Destination	Target	Notes
10.22.0.0/22	local	VPC local
0.0.0.0/0	vpce-fw-az-a	FW endpoint

C-WAN Attachment Subnet Route Table

Destination	Target	Notes
10.22.0.0/22	local	VPC local
0.0.0.0/0	vpce-fw-az-b	FW endpoint

C-WAN Segment Route Tables (at CNE level)

Segment	Prefix	Attachment	Type
Prod	10.20.0.0/21	vpc-prod-use1	propagated
Prod	10.30.0.0/21	vpc-prod-apse2	propagated
Non-Prod	10.21.0.0/21	vpc-nonprod-use1	propagated
Hybrid	10.0.0.0/8	dxgw-us	propagated
Hybrid	10.0.0.0/8	dxgw-apac	propagated
Shared Svc	TBD	vpc-shared-use1	propagated

Segment isolation enforced. Hybrid segment routes (DXGW) only reachable from Prod/Non-Prod via Inspection.

ap-southeast-2 Route Tables

lab-nonprod-apse2

Destination	Target	Notes
10.30.0.0/21	local	VPC local
0.0.0.0/0	cwan-att	

lab-inspection-apse2 — same pattern as us-east-1 (FW Subnet RT + C-WAN Attach RT)

lab-inspection-use1

Firewall Subnet Route Table

Destination	Target	Notes
10.22.0.0/22	local	VPC local
0.0.0.0/0	cwan-att	Return inspected traffic → C-WAN

C-WAN Attachment Subnet Route Table 1

Destination	Target	Notes
10.22.0.0/22	local	VPC local
0.0.0.0/0	vpce-fw-az-a	FW endpoint

C-WAN Attachment Subnet Route Table

Destination	Target	Notes
10.22.0.0/22	local	VPC local
0.0.0.0/0	vpce-fw-az-b	FW endpoint

Legend: Prod | Non-Prod | Inspection | Shared Svc | Hybrid (DXGW/VPN) | IBB | Firewall | send-to / send-via | CNE | ✗ No cross-geo leakage

E-W: ① ② ③ ④ ⑤ | N-S: Prod → Inspection → Hybrid → DXGW → On-Prem | Inbound: On-Prem → DXGW → Hybrid → Inspection → Prod

4.1 CNE Attachments

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Region	CNE Name	Attachment ID	Status
us-east-1	CNE-US-East	[TBP]	[TBP]
ap-southeast-2	CNE-APAC-Sydney	[TBP]	[TBP]
us-west-2	CNE-US-West	[TBP]	[TBP]

4.2 VPC & Inspection Resources

Region	VPC Name	CIDR	Type	Segment	FW Type
us-east-1	lab-prod-use1	10.20.0.0/21	Prod	Prod	—
us-east-1	lab-nonprod-use1	10.21.0.0/21	Non-Prod	Non-Prod	—
us-east-1	lab-inspection-use1	10.22.0.0/22	Inspection	Inspection	[AWS NFW / 3P]
us-east-1	lab-shared-use1	TBD	Shared Svc	Shared Svc	—
ap-southeast-2	lab-prod-apse2	10.30.0.0/21	Prod	Prod	—
ap-southeast-2	lab-inspection-apse2	10.32.0.0/22	Inspection	Inspection	[AWS NFW / 3P]
us-west-2	lab-prod-euc1	TBD	Prod	Prod	—
us-west-2	lab-nonprod-euc1	TBD	Non-Prod	Non-Prod	—
us-west-2	lab-inspection-euc1	TBD	Inspection	Inspection	[AWS NFW / 3P]

4.3 DXGW, Direct Connect & Megaport

DXGW	DXGW ID	ASN	Region	DX Conn	Megaport	Status
dxgw-us	[TBP]	[TBP]	us-east-1	[TBP]	[TBP]	Megaport ETA: Mar 20
dxgw-apac	[TBP]	[TBP]	ap-southeast-2	[TBP]	[TBP]	[TBP]

5. BGP Community Tagging & Route Propagation (need to get some clarity on the tagging plan)

5.1 BGP Community Tagging Plan

All routes must be tagged with Fiserv-required BGP communities per DR 2.5: below table is just a placeholder

Community Value	Meaning	Applied To
[TBP]	US origin prefix	All us-east-1 VPC routes
[TBP]	APAC origin prefix	All ap-southeast-2 VPC routes
[TBP]	US-West origin prefix	All us-west-2 VPC routes
[TBP]	On-prem DC prefix	Routes from DX connections

5.2 Route Propagation Rules ?? (More consensus between the team required e.g, engineering)

- **C-WAN Prod segment:** Prod VPC CIDRs propagate CNE-to-CNE within Prod segment only (isolated from Non-Prod)
- **C-WAN Non-Prod segment:** Non-Prod VPC CIDRs propagate within Non-Prod segment only (isolated from Prod)
- **C-WAN Shared Services segment:** Peers to both Prod and Non-Prod segments; Shared Services VPC routes visible in both
- **Inspection VPC attachment:** Receives return routes from Prod/Non-Prod VPCs; sends default or specific routes to attract inspection traffic
- **dxgw-us:** Receives aggregated US VPC routes from CNE-US-East; advertises to DX peers
- **dxgw-apac:** Receives Sydney VPC routes from CNE-APAC-Sydney; advertises to DX peers
- **dxgw-emea:** Receives Frankfurt VPC routes from CNE-USWEST-Oregon; advertises to DX peers
- **Cross-DXGW traffic:** Must traverse C-WAN backbone (not leak between DXGWs)

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6. Infrastructure as Code / Terraform

All lab infrastructure is managed via Terraform Enterprise with GitLab CI/CD runners.

Item	Details
GitLab Project-lab-repo	[GitLab repo URL]
State Backend	[S3, Dynamodb??]

Sample Module Layout:

```
mcbb3-lab-terraform/modules/
  cwan-core/           — C-WAN core network, segments, policies
  cne/                 — Cloud Network Edge per region
  vpc/                 — VPC, subnets, route tables
  inspection-vpc/      — Inspection VPC with firewall subnets
  network-firewall/    — AWS Network Firewall or NVA config
  dxgw/                — Direct Connect Gateway per region
  dx-connection/       — DX connections and VIFs
  ibb-segment/         — IBB C-WAN segment and policies
```

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7. Phased Deployment Runbook

The lab buildout is structured in sequential phases with dependencies and acceptance criteria mapped to Design Requirements (DRs).

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7.1 Phase 1: C-WAN Core Network & Segments

Dependencies: AWS account provisioned, Terraform Enterprise workspace configured, GitLab runner operational
DRs: DR 4.1, DR 2.10

- Deploy C-WAN core network with global policy document
- Create Prod, Non-Prod, Shared Services (peered to Prod + Non-Prod), and IBB segments across all three regions
- Deploy CNEs: CNE-US-East, CNE-APAC-Sydney, CNE-USWEST-Oregon
- Validate segment health and CNE state via console/API

Acceptance: All CNEs Available; four segments visible in all regions; Terraform state clean.

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7.2 Phase 2: VPC Deployment & Intra-Region Routing

Dependencies: Phase 1 complete
DRs: DR 1.1, DR 2.1, DR 2.10

- us-east-1: Deploy Prod (10.20.0.0/21), Non-Prod (10.21.0.0/21), Inspection (10.22.0.0/22), Shared Services (TBD)
- ap-southeast-2: Deploy Prod (10.30.0.0/21), Inspection (10.32.0.0/22)
- us-west-2: Deploy Prod (TBD), Non-Prod (TBD), Inspection (TBD)
- Attach Prod VPCs to Prod segment, Non-Prod VPCs to Non-Prod segment, Shared Services to Shared Services segment
- Configure Shared Services segment peering to both Prod and Non-Prod segments
- Attach Inspection VPCs as separate C-WAN attachments
- Deploy EC2 test instances in each VPC for connectivity validation
- Validate segment isolation: Prod cannot reach Non-Prod directly
- Validate Shared Services reachable from both Prod and Non-Prod via peering

Acceptance: Routes in correct segments; Prod/Non-Prod isolated; Shared Services peered to both; Inspection attachments visible.

7.3 Phase 3: Cross-Region C-WAN Routing

Dependencies: Phase 2 complete
DRs: DR 1.2, DR 2.2, DR 2.9

- Validate C-WAN propagates VPC routes between CNEs across all three regions
- Test cross-region connectivity: us-east-1 ap-southeast-2, us-east-1 us-west-2, ap-southeast-2 us-west-2
- Verify route tables via API/CLI (aws networkmanager get-network-routes)
- Measure cross-region latency baselines
- Confirm VPC routes do NOT leak to DXGW attachments

Acceptance: Full mesh ping across all three regions; route tables correct; no DXGW leakage.

7.4 Phase 4: Split DXGW & Direct Connect Integration

Dependencies: Phase 3 complete, Megaport circuit delivered (target: March 20)
DRs: DR 1.3, DR 1.4, DR 2.3, DR 1.9, DR 1.10, DR 2.5

- Create three regional DXGWs: dxgw-us, dxgw-apac, dxgw-emea
- Attach DXGWs to their respective CNEs
- Provision Megaport VXCs as hosted DX connections
- Establish BGP peering over transit VIFs
- Validate geo-affinity and DXGW failover
- Confirm cross-DXGW isolation (DR 1.12)
- Test BGP community tagging on advertised routes (DR 2.5)

Acceptance: On-prem reachable via correct regional DXGW; no cross-geo DX leakage.

7.5 Phase 5: Inspection VPC & Service Chaining

Dependencies: Phase 2 complete (can parallel Phases 3-4)
DRs: DR 1.6, DR 4.8, DR 4.9

- Deploy firewall in each region's Inspection VPC (start with AWS Network Firewall)
- Configure C-WAN send-to/send-via actions to steer east-west and north-south traffic through Inspection VPC
- Validate firewall logs show inspected flows
- Test traffic blackhole when firewall removed (no bypass path)
- Optionally test third-party NVA (CheckPoint, Palo Alto) as alternative in one region

Acceptance: All east-west/north-south traffic through Inspection VPC; firewall logs confirm; no bypass.

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7.6 Phase 6: IBB Architecture

Dependencies: Phase 4 complete
DRs: DR 1.5, DR 1.11, DR 1.12

- Deploy IBB C-WAN segment with dedicated route policies
- Test candidate IBB approaches:
  - **Option A:** C-WAN IBB segment with route policies restricting international traffic to approved paths
  - **Option B:** Accelerated Site-to-Site VPN tunnels for encrypted cross-region IBB underlay
  - **Option C:** Public VIF on existing DX for IBB overlay
- Validate international traffic paths and geopolitical routing constraints
- Document recommendation with pros/cons matrix

Acceptance: At least one IBB approach validated end-to-end with encryption; geopolitical constraints enforced.

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7.7 Phase 7: Observability & Telemetry

Dependencies: Phase 3+ (can parallel Phases 4-6)
DRs: DR 3.1, DR 3.3, DR 3.4

Target State (Cloud-Agnostic):

- ThousandEyes — Network path visualization, latency monitoring across all CSPs
- Splunk — Centralized log aggregation for flow logs, firewall logs, C-WAN events
- Selector AI — AIOps anomaly detection and automated root cause analysis

Interim Lab (AWS Native):

- CloudWatch metrics for C-WAN (bytes in/out, packet counts per attachment)
- VPC Flow Logs for all lab VPCs (S3 + CloudWatch Logs)
- Network Manager route analytics and topology visualization
- CloudWatch alarms for DX connection state changes
- AWS Network Firewall logs (alert + flow) from Inspection VPCs
- Document telemetry gaps and map to target state tool capabilities

Acceptance: Interim dashboards operational; gaps documented with target state mapping.

7.8 Phase 8: End-to-End Validation & Stress Testing

Dependencies: All previous phases complete
DRs: DR 2.7, DR 2.8, DR 2.10

- Full traffic flow matrix: every region to every region, every region to on-prem, all through Inspection VPCs
- Blackout simulation: kill primary DX, measure convergence
- Brownout simulation: inject latency/packet loss, observe routing behavior
- Multi-tenancy validation: secondary segment isolation
- iperf bandwidth tests across C-WAN backbone and DX connections
- Compile final test results against DR matrix

Acceptance: All DRs validated or documented with gaps/workarounds; final report compiled.

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8. Test Cases & Validation Matrix

Update Status as testing progresses. In Confluence, replace status text with `/status` macro for colored lozenges.

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Transport

DR	Description	Lab Test	Phase	Status
1.1	VPC-to-VPC same region	Ping Prod to Non-Prod in us-east-1	2	COMPLETED
1.2	VPC-to-VPC cross-region	Ping us-east-1 Prod to Sydney Prod	3	COMPLETED
1.3	VPC to on-prem (geo affinity)	US VPC reaches on-prem via dxgw-us only	4	NOT STARTED
1.4	VPC to on-prem (fallback)	Disable primary DX, verify alternate path	4	NOT STARTED
1.6	Network security enforcement	Traffic through Inspection VPC firewall	5	NOT STARTED
1.9	ECMP support	Multiple DX on same DXGW, verify load dist	4	NOT STARTED
1.12	Geopolitical routing avoidance	Verify APAC/EMEA traffic never routes via US DX	6	NOT STARTED

Routing

DR	Description	Lab Test	Phase	Status
2.1	Regional route propagation	VPC routes visible in regional CNE	2	COMPLETED
2.2	Cross-regional route propagation	VPC routes visible in remote CNEs	3	COMPLETED
2.5	BGP community tagging	Verify communities on DX-advertised routes	4	NOT STARTED
2.7	Sub-second convergence (blackout)	Kill DX link, measure convergence	8	NOT STARTED
2.9	End-to-end routing visibility	API/CLI route table dump all regions	3	NOT STARTED
2.10	Multi-tenancy (VRF-like)	Prod/Non-Prod segment isolation; no route leakage	2	COMPLETED

Observability

DR	Description	Lab Test	Phase	Status
3.1	Real-time flow telemetry	CloudWatch C-WAN metrics (interim); ThousandEyes (target)	7	IN-PROGRESS
3.3	Historical flow data	VPC Flow Logs to S3 (interim); Splunk (target)	7	IN-PROGRESS
3.4	Metrics push/pull /subscription	CloudWatch alarms (interim); Selector AI (target)	7	IN-PROGRESS

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Security & Service Chaining

DR	Description	Lab Test	Phase	Status
4.8	AWS ANF/NVA integration	Route traffic through firewall in Inspection VPC	5	COMPLETED
4.9	Service chaining	C-WAN send-to/send-via steers traffic to Inspection VPC	5	COMPLETED
	Security DMZ plan guiding Rules	Cloud DMZ Perimeter Design	5	IN-PROGRESS

9. Known Issues & Design Decisions

Design Decisions & Lab Deviations

ID	Decision	Rationale	Type	Impact
DD-001	Lab uses 3 regions (production has 11)	us-east-1, ap-southeast-2, us-west-2 cover all split DXGW boundaries	Deviation	Low
DD-002	NAT testing deferred (DR 4.2/4.3)	Production likely withdrawing NAT in favor of single GRT		Low
DD-003	Cloud-agnostic observability (ThousandEyes + Splunk + Selector AI)	Multi-cloud needs CSP-independent monitoring; CloudWatch as interim	Design Choice	Low
DD-004	Lab uses hosted DX (sub-1GE) vs production 10-100GE	Validates routing, not capacity	Deviation	Low
DD-005	CloudWatch interim for observability vs production TBD	Interim covers DRs; target validates multi-cloud readiness	Deviation	Low

Blockers

ID	Blocker	Impact	ETA
BLK-001	Megaport circuit delivery pending	Blocks Phase 4 (DX integration)	March 20, 2026